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BM2102 - Modelling and Analysis of
Physiological Systems

Electrical Properties of Branching
Dendrites

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1 Introduction

This report explores the time-independent (steady-state) electrical properties of branched cable models that represent dendritic trees in neurons. Neurons receive electrical signals through their dendrites, which branch out like trees. By treating each branch as a cable, we can mathematically describe how membrane voltage is distributed along the dendrite.

We consider a first-order branched cable: one parent branch that splits into two daughter branches. The membrane potential in each branch satisfies the steady-state cable equation, and by applying appropriate boundary and nodal conditions, we can solve for the voltage profile throughout the entire tree.

2 Question 1 – Deriving the System of Equations

Question 1

$$\begin{aligned}
 V_1(x) &= A_1 e^{-x} + B_1 e^x & 0 \leq x \leq L_1 \\
 V_{s1}(x) &= A_{s1} e^{-x} + B_{s1} e^x & L_1 \leq x \leq L_{s1} \\
 V_{s2}(x) &= A_{s2} e^{-x} + B_{s2} e^x & L_1 \leq x \leq L_{s2}
 \end{aligned}$$

Applying boundary conditions,

$$\left(\frac{dV_1}{dx} \right)_{x=0} = -(r_i \lambda_c) I_{app}$$

$$\frac{dV_1}{dx} = -A_1 e^{-x} + B_1 e^x$$

$$\left. \frac{dV_1}{dx} \right|_{x=0} = -A_1 + B_1 = -(r_i \lambda_c) I_{app}$$

$$A_1 - B_1 = (r_i \lambda_c) I_{app} \quad \text{--- (1)}$$

At terminal ends;

$$V_{s1}(L_{s1}) = 0$$

substituting $x = L_{s1}$

$$A_{s1} e^{-L_{s1}} + B_{s1} e^{L_{s1}} = 0 \quad \text{--- (2)}$$

$$V_{s2}(L_{s2}) = 0$$

substituting $x = L_{s2}$

$$A_{s2} e^{-L_{s2}} + B_{s2} e^{L_{s2}} = 0 \quad \text{--- (3)}$$

Figure 1: Handwritten derivation 1

Applying nodal conditions,

$$v_1(L_1) = v_2(L_1) = v_{22}(L_1)$$

$$A_1 e^{-L_1} + B_1 e^{L_1} = A_{21} e^{-L_1} + B_{21} e^{L_1}$$

$$A_1 e^{-L_1} + B_1 e^{L_1} - A_{21} e^{-L_1} - B_{21} e^{L_1} = 0 \quad \text{--- (4)}$$

$$A_{21} e^{-L_1} + B_{21} e^{L_1} = A_{22} e^{-L_1} + B_{22} e^{L_1}$$

$$A_{21} e^{-L_1} + B_{21} e^{L_1} - A_{22} e^{-L_1} - B_{22} e^{L_1} = 0 \quad \text{--- (5)}$$

Applying conservation of current;

$$\left. \frac{-1}{(r_i \lambda_c)_1} \frac{dv_1}{dx} \right|_{x=L_1} = \left. \frac{-1}{(r_i \lambda_c)_{21}} \frac{dv_{21}}{dx} \right|_{x=L_1} + \left. \frac{-1}{(r_i \lambda_c)_{22}} \frac{dv_{22}}{dx} \right|_{x=L_1}$$

$$\left. \frac{dv_1}{dx} \right|_{x=L_1} = -A_1 e^{-L_1} + B_1 e^{L_1}$$

$$\left. \frac{dv_{21}}{dx} \right|_{x=L_1} = -A_{21} e^{-L_1} + B_{21} e^{L_1}$$

$$\left. \frac{dv_{22}}{dx} \right|_{x=L_1} = -A_{22} e^{-L_1} + B_{22} e^{L_1}$$

$$\frac{-1}{(r_i \lambda_c)_1} (-A_1 e^{-L_1} + B_1 e^{L_1}) = \frac{-1}{(r_i \lambda_c)_{21}} (-A_{21} e^{-L_1} + B_{21} e^{L_1}) + \frac{-1}{(r_i \lambda_c)_{22}} (-A_{22} e^{-L_1} + B_{22} e^{L_1})$$

$$0 = \frac{-A_1 e^{-L_1}}{(r_i \lambda_c)_1} + \frac{B_1 e^{L_1}}{(r_i \lambda_c)_1} + \frac{A_{21} e^{-L_1}}{(r_i \lambda_c)_{21}} - \frac{B_{21} e^{L_1}}{(r_i \lambda_c)_{21}} + \frac{A_{22} e^{-L_1}}{(r_i \lambda_c)_{22}} - \frac{B_{22} e^{L_1}}{(r_i \lambda_c)_{22}}$$

--- (6)

Figure 2: Handwritten derivation 2

3 Question 2 – Matrix Multiplication Verification

The Matrix Equation

The six constraint equations from Question 1 can be written as the matrix equation:

$$\mathbf{Ax} = \mathbf{b}$$

Question 2

$$Ax = b$$

$$\begin{pmatrix} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & e^{-L_{21}} & e^{L_{21}} & 0 & 0 \\ 0 & 0 & 0 & 0 & e^{-L_{22}} & e^{L_{22}} \\ e^{-L_1} & e^{L_1} & -e^{-L_1} & -e^{L_1} & 0 & 0 \\ 0 & 0 & e^{-L_1} & e^{L_1} & -e^{-L_1} & -e^{L_1} \\ -\frac{e^{-L_1}}{(r_i \lambda_c)_1} & \frac{e^{L_1}}{(r_i \lambda_c)_1} & \frac{e^{-L_1}}{(r_i \lambda_c)_{21}} & -\frac{e^{L_1}}{(r_i \lambda_c)_{21}} & \frac{e^{-L_1}}{(r_i \lambda_c)_{22}} & -\frac{e^{L_1}}{(r_i \lambda_c)_{22}} \end{pmatrix} \begin{pmatrix} A_1 \\ B_1 \\ A_{21} \\ B_{21} \\ A_{22} \\ B_{22} \end{pmatrix} = \begin{pmatrix} (r_i \lambda_c)_1 I_{app} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} A_1 & -B_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & A_{21}e^{-L_{21}} & B_{21}e^{L_{21}} & 0 & 0 \\ 0 & 0 & 0 & 0 & A_{22}e^{-L_{22}} & B_{22}e^{L_{22}} \\ A_1e^{-L_1} & B_1e^{L_1} & -A_{21}e^{-L_1} & -B_{21}e^{L_1} & 0 & 0 \\ 0 & 0 & A_{21}e^{-L_1} & B_{21}e^{L_1} & -A_{22}e^{-L_1} & -B_{22}e^{L_1} \\ -\frac{A_1e^{-L_1}}{(r_i \lambda_c)_1} & \frac{B_1e^{L_1}}{(r_i \lambda_c)_1} & \frac{A_{21}e^{-L_1}}{(r_i \lambda_c)_{21}} & -\frac{B_{21}e^{L_1}}{(r_i \lambda_c)_{21}} & \frac{A_{22}e^{-L_1}}{(r_i \lambda_c)_{22}} & -\frac{B_{22}e^{L_1}}{(r_i \lambda_c)_{22}} \end{pmatrix} \begin{pmatrix} (r_i \lambda_c)_1 I_{app} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} A_1 - B_1 &= (r_i \lambda_c)_1 I_{app} \quad \text{--- (1)} \\ A_{21}e^{-L_{21}} + B_{21}e^{L_{21}} &= 0 \quad \text{--- (2)} \\ A_{22}e^{-L_{22}} + B_{22}e^{L_{22}} &= 0 \quad \text{--- (3)} \\ A_1e^{-L_1} + B_1e^{L_1} - A_{21}e^{-L_1} - B_{21}e^{L_1} &= 0 \quad \text{--- (4)} \\ A_{21}e^{-L_1} + B_{21}e^{L_1} - A_{22}e^{-L_1} - B_{22}e^{L_1} &= 0 \quad \text{--- (5)} \\ -\frac{A_1e^{-L_1}}{(r_i \lambda_c)_1} + \frac{B_1e^{L_1}}{(r_i \lambda_c)_1} + \frac{A_{21}e^{-L_1}}{(r_i \lambda_c)_{21}} - \frac{B_{21}e^{L_1}}{(r_i \lambda_c)_{21}} + \frac{A_{22}e^{-L_1}}{(r_i \lambda_c)_{22}} - \frac{B_{22}e^{L_1}}{(r_i \lambda_c)_{22}} &= 0 \quad \text{--- (6)} \end{aligned}$$

Figure 3: Handwritten derivation 3

4 Question 3 – Solving for the Coefficients

MATLAB Code

The full setup and solution are as follows:

```

1 % electrical constants and derived quantities for typical
  mammalian dendrite
2
3 % Dimensions of compartments
4 d1 = 75e-4; % cm (parent branch diameter)
5 d21 = 30e-4; % cm (daughter branch 21 diameter)
6 d22 = 15e-4; % cm (daughter branch 22 diameter)

```

```

7
8 l1 = 1.5;           % dimensionless electrotonic length (parent)
9 l21 = 3.0;          % dimensionless electrotonic length (daughter 21)
10 l22 = 3.0;          % dimensionless electrotonic length (daughter 22)
11
12 % Electrical properties
13 Rm = 6e3;           % Ohms cm^2 (membrane resistance)
14 Rc = 90;            % Ohms cm (cytoplasmic resistance)
15 Rs = 1e6;           % Ohms (seal resistance)
16
17 % Derived constant
18 c1 = 2*(Rc*Rm)^(1/2)/pi;
19
20 % Axial resistance * lambda_c for each branch
21 rl1 = c1 * d1^(-3/2); % Ohms
22 rl21 = c1 * d21^(-3/2); % Ohms
23 rl22 = c1 * d22^(-3/2); % Ohms
24
25 % Applied current
26 iapp = 1e-9;        % Amps (1 nA)
27
28 % Coefficient matrix A (6x6)
29 A = [1,              -1,              0,              0,              0,
30       0,              0;
31       0,              0,      exp(-l21),      exp(l21),              0,
32       0,              0;
33       0,              0,              0,              0,      exp(-l22),
34       exp(l22);
35       exp(-l1),      exp(l1), -exp(-l1), -exp(l1),              0,
36       0;
37       0,              0,      exp(-l1),      exp(l1),      -exp(-l1),      -exp
38       (l1);
39       -exp(-l1),      exp(l1), rl1*exp(-l1)/rl21, -rl1*exp(l1)/rl21,
40       ...
41       rl1*exp(-l1)/rl22, -rl1*exp(l1)/rl22];
42
43 % Right-hand side vector b (column vector)
44 b = [rl1*iapp; 0; 0; 0; 0; 0];
45
46 disp('== Setup complete. rl1, rl21, rl22 values: ==');
47 fprintf('rl1 = %.4f Ohms\n', rl1);
48 fprintf('rl21 = %.4f Ohms\n', rl21);
49 fprintf('rl22 = %.4f Ohms\n', rl22);
50
51 % Solve for the 6 coefficients
52 x = A\b;
53 disp('Coefficients [A1 B1 A21 B21 A22 B22]:');
54 disp(x)

```

Listing 1: MATLAB code for solving for coefficients (Question 3).

Results

Running the above code gives the following coefficient values:

```

=== Setup complete. r11, r121, r122 values: ===
r11  = 720253.0529 Ohms
r121 = 2847050.1737 Ohms
r122 = 8052673.9367 Ohms
=== Question 3: Solved coefficients x = [A1; B1; A21; B21; A22; B22] ===
    0.0007
    0.0000
    0.0011
   -0.0000
    0.0011
   -0.0000

```

Figure 4: Results of X values

Table 1: Solved coefficients from the linear system $\mathbf{Ax} = \mathbf{b}$.

Coefficient	Value (approx.)
A_1	7×10^{-4}
B_1	≈ 0
A_{21}	1.1×10^{-3}
B_{21}	≈ 0
A_{22}	1.1×10^{-3}
B_{22}	≈ 0

The near-zero values of B_1 , B_{21} , and B_{22} indicate that the growing exponential components are essentially absent, which means the voltage decays as we move further from the current injection site.

5 Question 4 – Steady-State Voltage Profile

MATLAB Code

```

1 % (Run after Question 3 code -- variable x must be defined)
2
3 % Create x-axis vectors for each branch
4 y1 = linspace(0, 11, 20); % parent branch: X in [0, L1]
5 y21 = linspace(11, 121, 20); % daughter branch 21: X in [L1,
   L21]
6 y22 = linspace(11, 122, 20); % daughter branch 22: X in [L1,
   L22]
7
8 % Voltage profiles using the solved coefficients
9 v1 = x(1)*exp(-y1) + x(2)*exp(y1);
10 v21 = x(3)*exp(-y21) + x(4)*exp(y21);
11 v22 = x(5)*exp(-y22) + x(6)*exp(y22);
12
13 % Plot all three branches on the same axes
14 plot(y1, v1, 'y-', y21, v21, 'r-', y22, v22, 'b-');

```

```

15 xlabel('X (dimensionless)');
16 ylabel('V (Volts)');
17 title('Steady-State Voltage - E5');
18 legend('Parent branch V_1', 'Daughter branch V_{21}', 'Daughter
    branch V_{22}');
19 grid on;

```

Listing 2: MATLAB code for plotting the steady-state voltage profile .

Resulting Plot

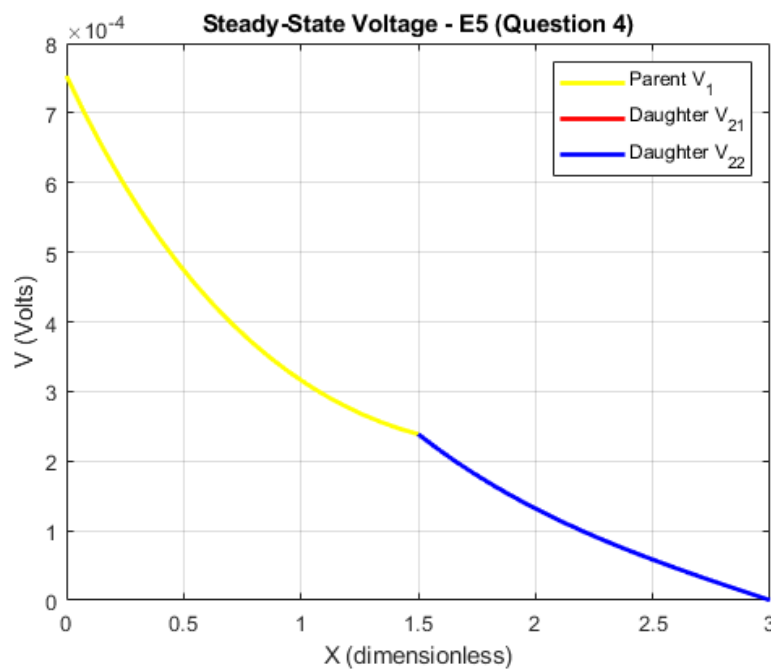


Figure 5: Steady state voltage profile

Observation

Looking at the plot, the red line (V_{21}) and blue line (V_{22}) completely overlap and they are indistinguishable. This means that both daughter branches have exactly the same steady-state voltage profile.

This result makes intuitive sense for two reasons. First, both daughter branches share the same electrotonic length and the same boundary condition (killed end at the tip). Second, the current conservation at the branch node distributes current into each daughter branch in proportion to their conductances. Since both daughters have identical terminal conditions and lengths, the voltage distribution must also be identical.

Even though $d_{21} \neq d_{22}$ (they have different diameters), the solved coefficients $A_{21} = A_{22}$ and $B_{21} = B_{22}$, confirming that the profiles are the same.

6 Question 5 – Different Boundary Conditions

6.1 Part (a)

$$\left. \frac{dV_1}{dX} \right|_{X=0} = -(r_i \lambda_c)_1 I_{app}, \quad \left. \frac{dV_{21}}{dX} \right|_{X=L_{21}} = 0, \quad V_{22}(L_{22}) = 0$$

```

1 % Start fresh from the original A and b defined in Question 3
2 A1 = A;
3 A1(2,:) = [0, 0, -exp(-121), exp(121), 0, 0]; % sealed end on
   V21
4 % b remains the same (current injected at X=0)
5 x = A1\b;
6
7 % Plot
8 y1 = linspace(0, 11, 20);
9 y21 = linspace(11, 121, 20);
10 y22 = linspace(11, 122, 20);
11
12 v1 = x(1)*exp(-y1) + x(2)*exp(y1);
13 v21 = x(3)*exp(-y21) + x(4)*exp(y21);
14 v22 = x(5)*exp(-y22) + x(6)*exp(y22);
15
16 plot(y1, v1, 'y-', y21, v21, 'r-', y22, v22, 'b-');
17 xlabel('X (dimensionless)');
18 ylabel('V (Volts)');
19 title('Steady-State Voltage - Case (a)');
20 legend('V_1', 'V_{21}', 'V_{22}');
21 grid on;

```

Listing 3: MATLAB code for boundary condition (a).

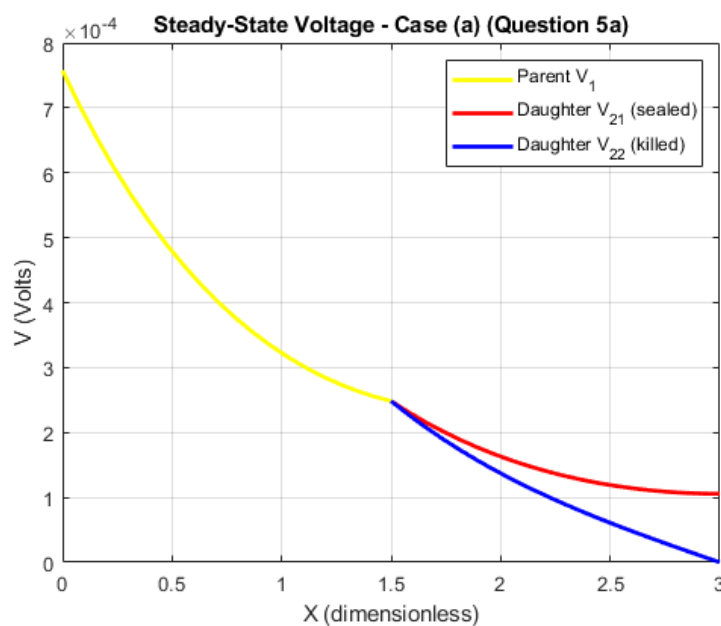


Figure 6: Steady-state voltage profile for boundary condition (a).

6.2 Part (b)

$$\left. \frac{dV_1}{dX} \right|_{X=0} = -(r_i \lambda_c)_1 I_{app}, \quad \left. \frac{dV_{21}}{dX} \right|_{X=L_{21}} = 0, \quad \left. \frac{dV_{22}}{dX} \right|_{X=L_{22}} = 0$$

```

1 % Continue from Part (a) -- A1 already has row 2 modified
2 A1(3,:) = [0, 0, 0, 0, -exp(-122), exp(122)]; % sealed end on
   V22
3 x = A1\b;
4
5 y1 = linspace(0, 11, 20);
6 y21 = linspace(11, 121, 20);
7 y22 = linspace(11, 122, 20);
8
9 v1 = x(1)*exp(-y1) + x(2)*exp(y1);
10 v21 = x(3)*exp(-y21) + x(4)*exp(y21);
11 v22 = x(5)*exp(-y22) + x(6)*exp(y22);
12
13 plot(y1, v1, 'y-', y21, v21, 'r-', y22, v22, 'b-');
14 xlabel('X (dimensionless)');
15 ylabel('V (Volts)');
16 title('Steady-State Voltage - Case (b)');
17 legend('V_1', 'V_{21}', 'V_{22}');
18 grid on;

```

Listing 4: MATLAB code for boundary condition (b).

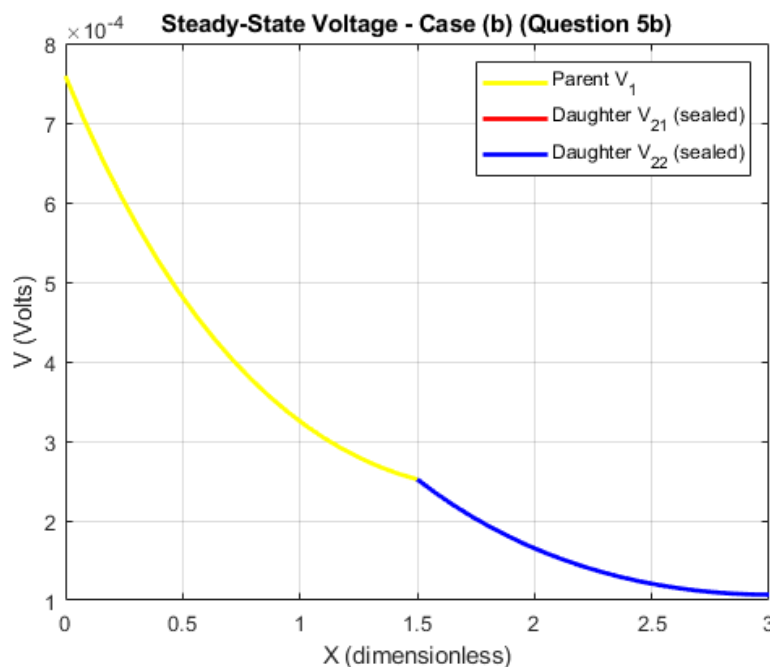


Figure 7: Steady-state voltage profile for boundary condition (b).

6.3 Part (c)

$$\left. \frac{dV_1}{dX} \right|_{X=0} = 0, \quad \left. \frac{dV_{21}}{dX} \right|_{X=L_{21}} = (r_i \lambda_c)_{21} I_{app}, \quad V_{22}(L_{22}) = 0$$

```

1 % Start with the original A matrix
2 A2 = A;
3 A2(2,:) = [0, 0, -exp(-121), exp(121), 0, 0]; % sealed end at
   tip of V21
4
5 % Modify b: no current at parent end; current injected at tip of
   V21
6 b2 = b;
7 b2(1) = 0; % sealed parent end: dV1/dX|_{X=0} = 0
8 b2(2) = r121 * iapp; % current injected at tip of V21
9
10 x = A2\b2;
11
12 y1 = linspace(0, 11, 20);
13 y21 = linspace(11, 121, 20);
14 y22 = linspace(11, 122, 20);
15
16 v1 = x(1)*exp(-y1) + x(2)*exp(y1);
17 v21 = x(3)*exp(-y21) + x(4)*exp(y21);
18 v22 = x(5)*exp(-y22) + x(6)*exp(y22);
19
20 plot(y1, v1, 'y-', y21, v21, 'r-', y22, v22, 'b-');
21 xlabel('X (dimensionless)');
22 ylabel('V (Volts)');
23 title('Steady-State Voltage - Case (c)');
24 legend('V_1', 'V_{21}', 'V_{22}');
25 grid on;

```

Listing 5: MATLAB code for boundary condition (c).

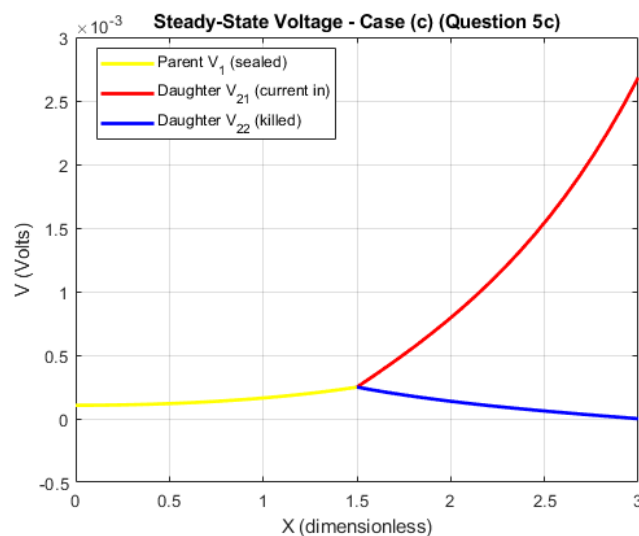


Figure 8: Steady-state voltage profile for boundary condition (c).

6.4 Part (d)

$$\left. \frac{dV_1}{dX} \right|_{X=0} = 0, \quad \left. \frac{dV_{21}}{dX} \right|_{X=L_{21}} = (r_i \lambda_c)_{21} I_{app}, \quad \left. \frac{dV_{22}}{dX} \right|_{X=L_{22}} = (r_i \lambda_c)_{22} I_{app}$$

```

1 % b2 already has b2(1)=0 and b2(2)=r121*iapp from Part (c)
2 b2(3) = r122 * iapp;      % current also injected at tip of V22
3
4 % Use A1 from Part (b) -- both daughter ends are sealed (no
  current flows out)
5 x = A1\b2;
6
7 y1 = linspace(0, 11, 20);
8 y21 = linspace(11, 121, 20);
9 y22 = linspace(11, 122, 20);
10
11 v1 = x(1)*exp(-y1) + x(2)*exp(y1);
12 v21 = x(3)*exp(-y21) + x(4)*exp(y21);
13 v22 = x(5)*exp(-y22) + x(6)*exp(y22);
14
15 plot(y1, v1, 'y-', y21, v21, 'r-', y22, v22, 'b-');
16 xlabel('X (dimensionless)');
17 ylabel('V (Volts)');
18 title('Steady-State Voltage - Case (d)');
19 legend('V_1', 'V_{21}', 'V_{22}');
20 grid on;

```

Listing 6: MATLAB code for boundary condition (d).

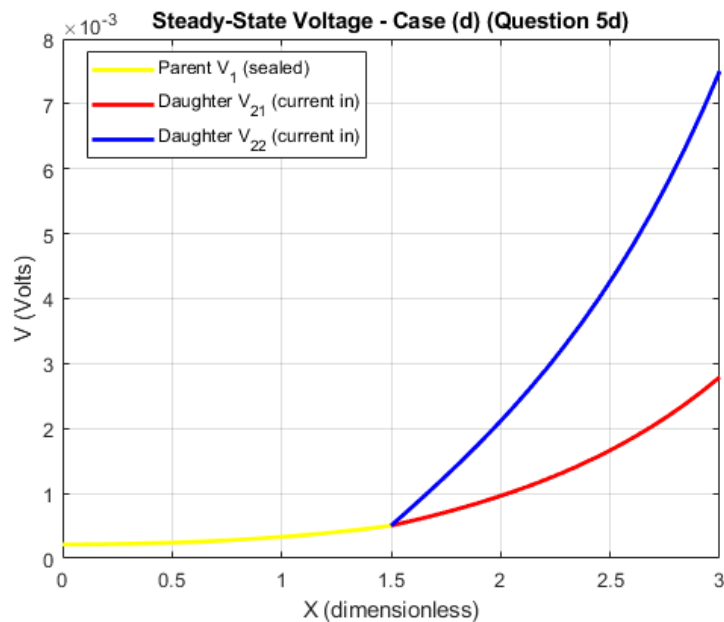


Figure 9: Steady-state voltage profile for boundary condition (d).

7 Question 5 – Physical Meaning of the Positive Gradient

A positive value of dV_{21}/dX or dV_{22}/dX at the terminal end simply means that current is being injected into the cable at that tip. This is because in cable theory, current and voltage gradient are related by:

$$I = \frac{-1}{r_i \lambda_c} \frac{dV}{dX}$$

When $dV/dX > 0$, this gives a negative current, meaning current flows into the branch at that point – just like a small electrode delivering a stimulus at the tip of a daughter branch.

In contrast, a zero gradient (sealed end) means no current flows at all, and a zero voltage (killed end) acts as a current sink that absorbs everything reaching it.

8 Question 6 – Effect of Equal Daughter Diamters

MATLAB Code

```

1 % Question 6 -- recalculate with d21 = d22 = 47.2470e-4 cm (Rall's
  branching rule)
2
3 % Dimensions
4 d1 = 75e-4;           % cm (parent -- unchanged)
5 d21 = 47.2470e-4;     % cm (daughter 21 -- changed)
6 d22 = d21;           % cm (daughter 22 = daughter 21)
7
8 l1 = 1.5;
9 l21 = 3.0;
10 l22 = 3.0;
11
12 % Electrical properties (unchanged)
13 Rm = 6e3;   Rc = 90;   Rs = 1e6;
14
15 c1 = 2*(Rc*Rm)^(1/2)/pi;
16 r11 = c1 * d1^(-3/2);
17 r121 = c1 * d21^(-3/2);
18 r122 = c1 * d22^(-3/2);
19
20 iapp = 1e-9;
21
22 % Rebuild A and b from scratch
23 A = [1,          -1,          0,          0,          0,
      0,          0;
24      0,          0,  exp(-l21),  exp(l21),  0,
      0;
25      0,          0,          0,          0,  exp(-l22),
  exp(l22);
26      exp(-l1),  exp(l1), -exp(-l1), -exp(l1),  0,
      0;
27      0,          0,  exp(-l1),  exp(l1),  -exp(-l1),  -exp
  (l1)];

```

```

28     -exp(-l1),    exp(l1), r11*exp(-l1)/r121, -r11*exp(l1)/r121,
29     ...
30         r11*exp(-l1)/r122, -r11*exp(l1)/r122];
31 b = [r11*iapp; 0; 0; 0; 0; 0];

```

Listing 7: Full MATLAB code for equal daughter diameters.

8.1 Part (b)

```

1 % Question 6 - Part (b): sealed ends on both daughter branches
2 A3 = A;
3 A3(2,:) = [0, 0, -exp(-l21), exp(l21), 0, 0];    % sealed end on
4           V21
5 A3(3,:) = [0, 0, 0, 0, -exp(-l22), exp(l22)];    % sealed end on
6           V22
7 x = A3\b;
8
9 y1 = linspace(0, l1, 20);
10 y21 = linspace(l1, l21, 20);
11 y22 = linspace(l1, l22, 20);
12
13 v1 = x(1)*exp(-y1) + x(2)*exp(y1);
14 v21 = x(3)*exp(-y21) + x(4)*exp(y21);
15 v22 = x(5)*exp(-y22) + x(6)*exp(y22);
16
17 plot(y1, v1, 'y-', y21, v21, 'r-', y22, v22, 'b-');
18 xlabel('X (dimensionless)');
19 ylabel('V (Volts)');
20 title('Q6 Steady-State Voltage - Case (b), d21=d22=47.2470e-4 cm'
21       );
22 legend('V_1', 'V_{21}', 'V_{22}');
23 grid on;

```

Listing 8: MATLAB code for Question 6 Part (b).

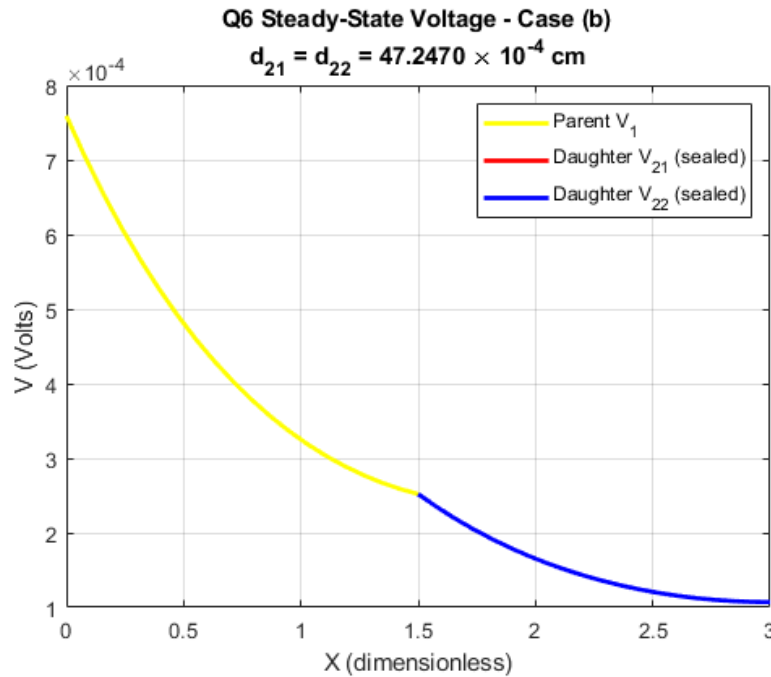


Figure 10: Steady-state voltage for Question 6 Part (b) with equal daughter diameters

8.2 Part (d)

```

1 % Question 6 - Part (d): current injected at both daughter tips;
  sealed parent end
2 A3 = A;
3 A3(2,:) = [0, 0, -exp(-121), exp(121), 0, 0]; % sealed end (
  current in) V21
4 A3(3,:) = [0, 0, 0, 0, -exp(-122), exp(122)]; % sealed end (
  current in) V22
5
6 b3 = b;
7 b3(1) = 0; % sealed parent: no current at X=0
8 b3(2) = r121 * iapp; % current injected at tip of V21
9 b3(3) = r122 * iapp; % current injected at tip of V22
10
11 x = A3\b3;
12
13 y1 = linspace(0, 11, 20);
14 y21 = linspace(11, 121, 20);
15 y22 = linspace(11, 122, 20);
16
17 v1 = x(1)*exp(-y1) + x(2)*exp(y1);
18 v21 = x(3)*exp(-y21) + x(4)*exp(y21);
19 v22 = x(5)*exp(-y22) + x(6)*exp(y22);
20
21 plot(y1, v1, 'y-', y21, v21, 'r-', y22, v22, 'b-');
22 xlabel('X (dimensionless)');
23 ylabel('V (Volts)');
24 title('Q6 Steady-State Voltage - Case (d), d21=d22=47.2470e-4 cm'
  );

```

```

25 legend('V_1', 'V_{21}', 'V_{22}');
26 grid on;

```

Listing 9: MATLAB code for Question 6 Part (d).

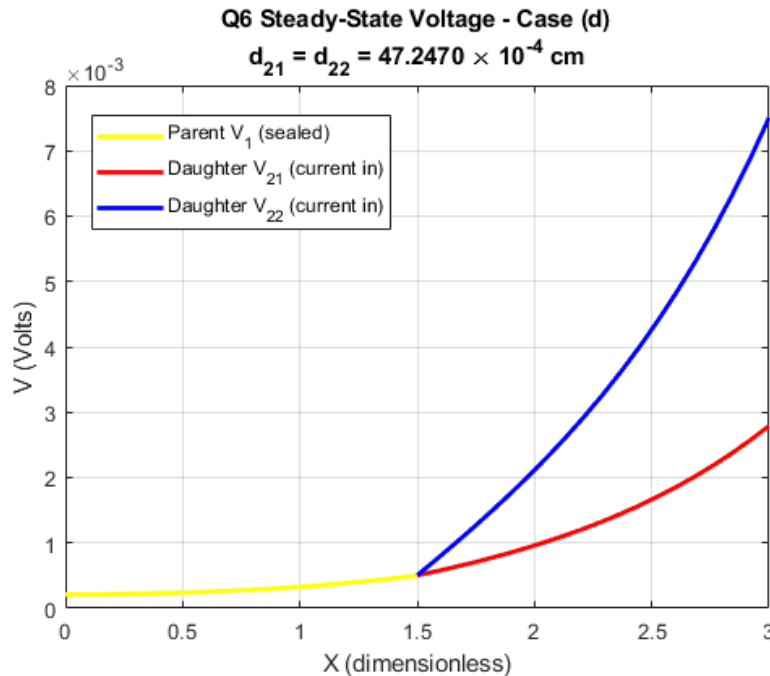


Figure 11: Steady-state voltage for Question 6 Part (d) with equal daughter diameters

Observations for Question 6

When the daughter branch diameters are made equal, two things stand out. First, both daughter branches show exactly the same voltage profile since they now have the same diameter, length, and boundary conditions. Second, this specific diameter value satisfies Rall's branching rule, which means the branch point becomes electrically invisible to the parent branch. The parent simply sees the two daughters as one continuous cable, so the voltage decays smoothly without any disturbance at the junction.

9 Conclusion

In this assignment, we modelled a branched dendritic cable and studied how the steady-state voltage behaves under different boundary conditions. The voltage in each branch follows an exponential shape, and by solving a linear system built from the boundary and nodal conditions, we can find the exact profile throughout the tree. Changing the boundary conditions at the tips has a significant effect on the voltage shape. Sealed ends keep the voltage from dropping to zero, while injecting current at the tips drives the voltage upward toward the branch point. Finally, when the daughter diameters satisfy Rall's branching rule, the branch point causes no disruption to the signal, and both daughters carry identical voltage profiles.